



Physicist-Engineer Nanotechnology and Quantum Applications Specialization Laboratory Measurement 1:

Neuromorphic Electronics

Abstract. According to an analysis, by 2030, information technology will account for more than 20% of the world's total energy consumption due to the rapid growth in stored and processed data and the rise of artificial intelligence. To prevent data centers from consuming the world's electricity production, in addition to developing artificial intelligence algorithms, new energy-efficient hardware devices are also needed that better follow the structure of neural networks and even handle data storage and computing in one place. This measurement session provides insight into the emerging field of research known as neuromorphic electronics. The measurement description presents the basics of the field, introducing new types of memories, known as memristors, which can behave as artificial synapses or artificial neurons. During the measurement task, you will examine volatile vanadium dioxide (VO_2) memristors, which can be used to create oscillatory circuits or even oscillating neural networks or artificial neurons.

Warning. During the measurement, you will study the temperature-dependent phase transition of the samples under investigation, during which you must not touch the hot surfaces. In addition, you will be working with measuring systems that operate at low voltages, which pose a greater risk to the samples under investigation than to the persons performing the measurements. In order to protect the nanofabricated circuits studied during the measurement, it is very important to check the drive signal shapes before the actual measurement, as signal shapes with inappropriate amplitude or duration can cause damage to the device under test. Discharges caused by electrostatically charged bodies can also cause damage to the devices, so ESD (electrostatic discharge) protection must be used.

As a general rule, always ensure the safety of yourself and others during measurements. If you are unsure about anything, ask the lab supervisors for help and always follow their warnings and instructions.

Introduction

1 Need for Novel Hardware Architectures in Information Technologies

In general, it can be stated that artificial neural networks, which form the basis of artificial intelligence, are capable of solving very complex tasks, but running them on traditional digital computers requires enormous computing power, as a very large number of neural output values and synaptic weights must be multiplied and added together. In addition, data storage and operations are performed in physically separate units, which means that data transfer between memory and processor also becomes a bottleneck.

With its 175 million parameters ChatGPT claims that it needs to perform 10-20 trillion (billion times billion) calculations to answer a single question. This computational demand is also significant in terms of energy consumption. According to a prediction, by 2030,

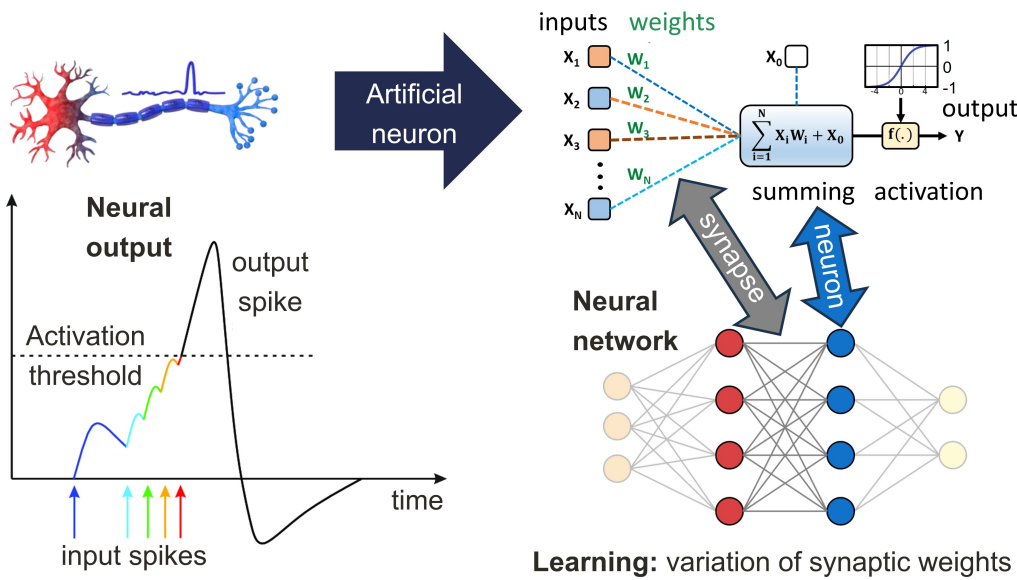


Figure I.1.1. Left: illustration of biological neurons. In a biological neural network, a neuron collects information from other neurons through its various synapses, with the synaptic connections having different strengths (different synaptic weights). The information arriving via the synapses is added up, and if the total strength of the input neural pulses exceeds an activation threshold, the neuron emits an output pulse. Learning schemes are associated with changes in synaptic weights, according to the relevance of the actual piece of information. **Right: illustration of artificial neural networks.** In the most common artificial neural networks a neuron also collects information from its inputs, which are usually the output values of other neurons (x_i). These are multiplied by the synaptic weights, and added together also using an offset value, $\sum_i x_i w_i + x_0$. Finally a nonlinear activation function, $f(\cdot)$ is applied, which outputs a certain value (like -1) if the sum is well below a threshold, and changes to another value (like $+1$) if the *summed information* exceeds the activation threshold. The simplest feed-forward neural networks (bottom) contain an input layer and an output layer (left and right yellow circles), as well as additional hidden neural layers (red and blue circles). All neurons between two neighboring layers (e.g. red and blue) are connected by artificial synapses. Here, the learning scheme is also related to the change in weights via the so-called backpropagation algorithm.

more than 20% of the world's total energy consumption will be required by information technology based on artificial intelligence [1]. For this reason, it is not enough to develop algorithms and device networks capable of solving more and more complex tasks in order to further advance artificial intelligence. It is at least as important to be able to perform the underlying calculations quickly and energy-efficiently with new types of hardware devices that are better aligned with the structure of neural networks and can handle data storage and computation in one place.

The biological nervous system provides inspiration for the development of this new, so-called *neuromorphic* hardware. Just think about how incredibly efficient the human brain is, consuming only about 30 W of energy! Fig. 1.1.1 briefly summarizes how artificial neural networks are inspired by the biological nervous system. Such neural networks are already easy to implement in software, but neuromorphic electronics aims to implement artificial neural networks in energy-efficient hardware.

2 Non-volatile memristors as artificial synapses

The possible building blocks of such neuromorphic hardware are so-called memristors, i.e., resistors with memory. These are physical systems whose resistance can be tuned with high voltage and read without information loss at low voltage. Physically, a memristor is implemented by placing a special, originally insulating layer between two metal electrodes. In many applications this insulating layer is a metal oxide, like Ta_2O_5 or HfO_x . With sufficiently high electrical voltage, we can move oxygen ions out of a certain region, thus turning this oxygen-deficient region into a conductor (see the illustration in the bottom inset of Fig. 1.2.1a). With this process we can form a well-conducting metallic filament between the two electrodes. At opposite voltage polarity, however, the oxygen ions return to this filamentary region (see the illustration in the top inset of Fig. 1.2.1a) thus blocking conduction through the filament. This reproducible process is illustrated by the current-voltage ($I(V)$) characteristic of a Ta_2O_5 memristor shown in Fig. 1.2.1a. Here, switching occurs between a certain low resistance state (LRS) and a certain high resistance state (HRS), but the actual resistance of the HRS and LRS, i.e., the corresponding filament diameter, can be adjusted by the amplitude and duration of the applied voltage signal. Furthermore, at sufficiently low voltages (below the positive and negative switching thresholds), the system maintains its state, which means that the memristor behaves as a constant resistance R_M with an almost linear $I(V)$ curve, $I = V/R_M$. In this low voltage range, the resistance state can be read without changing the resistance. Such memristors are analog memories whose low-voltage $I(V)$ curve slope can be adjusted practically to arbitrary values (see Fig. 1.2.1b). Accordingly, their conductance $G_M = 1/R_M$ can be used efficiently for the hardware implementation of synaptic weights in artificial neural networks.

In order to use memristors in information technology, they must be arranged in a network. This is surprisingly easy to achieve: all that is needed is to place an insulating layer suitable for forming memristors between the blue bottom and red top metal electrodes, as illustrated in Fig. 1.2.2. When voltage is applied between the top electrode i and the bottom electrode j , a memristor contact can be formed at the crosspoint of these electrodes, whose conductance G_{ji} can be set to the desired value. These memristor contacts are represented by the brown filaments in Fig. 1.2.2. It should be noted that once the conductance matrix G_{ji} has been programmed into this so-called crossbar memristor array, the voltage vector V_i applied to the upper electrodes results in a current output vector I_j at the lower electrodes, which is simply the matrix-vector product of the input voltage vector and the memristor conductance matrix, $I_j = \sum_i G_{ji} \cdot V_i$. It should be noted that in a neural net-

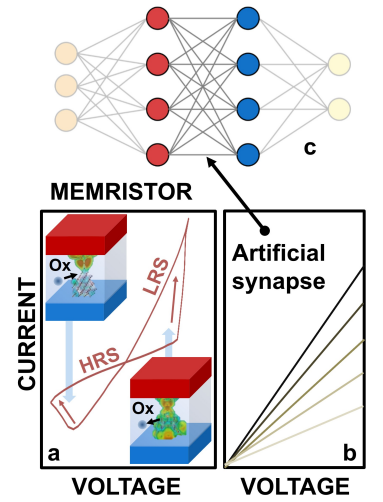


Figure 1.2.1. Memristors as artificial synapses. (a) Illustrative $I(V)$ curve of a Ta_2O_5 memristor. At a positive threshold the high resistance state (HRS) is switched to a low resistance state (LRS) due to the removal of oxygen ions from the active filamentary region, while at opposite polarity backswitching occurs (see the illustrations in the insets). (b) A memristor is an artificial synapse, as the slope of its low-voltage $I(V)$ curve can be adjusted almost arbitrarily with the appropriate voltage manipulation. (c) This property can be used in neural networks for the hardware implementation of synaptic weights.

work, such as the one shown in Fig. I.1.1, the input vector of the blue neural layer i_j is given by: $i_j = \sum_i w_{ji} \cdot x_i$, where x_i is the output vector of the red neural layer and w_{ji} is the weight matrix of the gray synaptic connections between the two layers. If there are N neurons in both layers, evaluating this input vector requires N^2 multiplication operations in the software. However, if the synaptic connections are implemented in hardware using a memristive crossbar array, this matrix-vector multiplication calculation is performed by the memristive hardware in a single step. This is a significant advantage in terms of energy efficiency and computational speed. Non-volatile memristors, which function as artificial synapses, have numerous applications in the field of neuromorphic computing, but the hardware-level single-step matrix-vector multiplication presented here is undoubtedly one of their most promising applications. Nowadays, this is no longer just a theoretical concept, but with the construction of networks of tens of millions of memristors, it is possible to build artificial neural networks that are capable of performing complex image recognition or other computational tasks with high accuracy and astonishing energy efficiency [2, 3].

3 Volatile memristors as artificial neurons

In the above example, non-volatile memristors functioned as *artificial synapses* and essentially served as hardware accelerators for matrix-vector multiplication calculations. However, other types of memristors can also behave as *artificial neurons*. Such memristors are generally volatile, which means that at a given voltage they switch from their original high-resistance state to a low-resistance state, but at the same voltage polarity they switch back to their original high-resistance state with a hysteresis once the voltage is lowered. Accordingly, at zero bias, always the high-resistance state is realized.

This operation can be performed, for example, with vanadium dioxide (VO_2) memristors, which are also the candidates for this measurement session. VO_2 is a special material system that exhibits an insulator-metal transition at a temperature around $T = 68^\circ\text{C}$, as shown by its temperature-dependent resistance in Fig. I.3.1a. The same insulator-metal transition can also be triggered by voltage when applied to a nanodevice in which two metal electrodes are connected by a narrow VO_2 region (Fig. I.3.1b). In this case, the phase transition is primarily caused by self-heating due to the voltage, but nonlinear conduction phenomena may also play a role [4]. This voltage-induced resistance switching is illustrated by the $I(V)$ curves in Fig. I.3.1c. In this case, the driving voltage V_{drive} is ramped by a

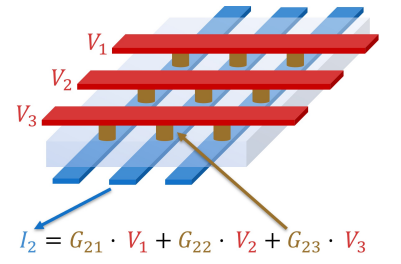


Figure I.2.2. Memristive crossbar arrays. The brown memristor contacts are formed at the intersection of the upper red and lower blue electrodes. Such a crossbar array implements synaptic connections between two adjacent layers of a neural network. The memristor conductance matrix, G_{ji} , is programmed during the training of the neural network. Subsequently, during inference, the synaptic weights are fixed, and the memristive crossbar array performs matrix-vector multiplication in a single hardware operation according to Ohm's law and Kirchoff's rules.

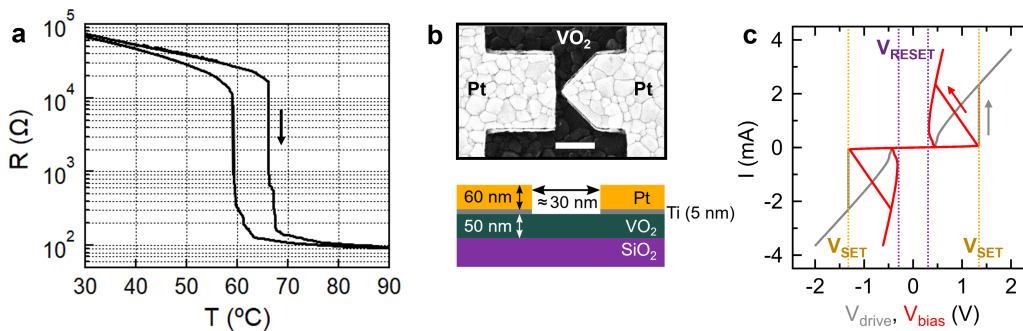


Figure I.3.1. Characteristic traits of VO_2 nanogap memristors. Temperature-dependent resistance of a nanogap memristor taken from Ref. 4. (b) Top: Scanning electron microscopy (SEM) image of the active region of a representative memristor studied in this laboratory exercise. Scale bar: 500 nm. Bottom: schematic material structure of the investigated memristors. (c) Drive voltage (gray) and bias voltage (red) -dependent $I(V)$ characteristics of a VO_2 memristor.

voltage generator, and this signal is fed to the VO_2 device and a series-connected resistor R_S . Fig. I.3.1 shows the $I(V)$ curve as a function of both V_{drive} (gray curve) and the bias voltage, $V_{\text{bias}} = V_{\text{drive}} - I \cdot R_S$, which drops across the VO_2 device itself (red curve). The latter curve shows the so-called set voltage, where the VO_2 device switches from HRS to LRS, and the reset voltage, where it switches back from LRS to HRS. It should be noted that when the switching occurs, the resistance of the VO_2 device changes, and therefore V_{bias} also changes due to the voltage division with the series-connected resistor. Such $I(V)$ curves (like Fig. I.3.1c) and temperature dependent resistance curves (like Fig. I.3.1a) will be investigated during the first part of the measurement session.

Circuits containing as few as two VO_2 memristors are capable of reproducing various types of biological neural spiking patterns [5]. Fig. I.3.2 shows the simplest case: an artificial neuron circuit that emits a single neural spike when the input exceeds the proper threshold value.

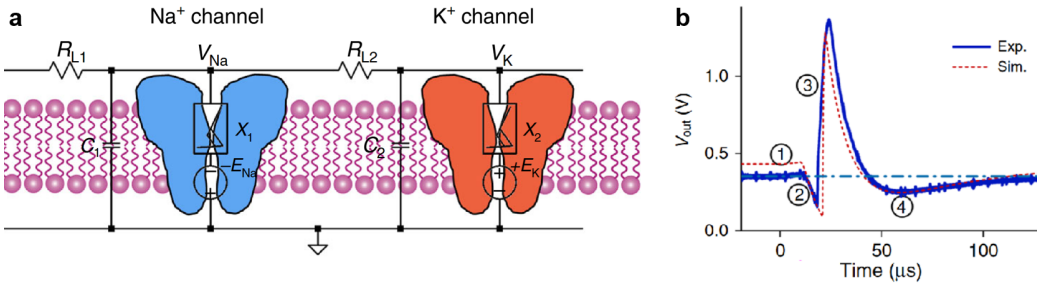


Figure I.3.2. Bio-inspired circuits. (a) Schematics of an artificial neuronal circuit made of a pair of VO_2 memristors, several R , C elements and additional DC voltage sources. The background color figure illustrates that this artificial neural circuit performs similar operation to the Na^+ and K^+ channels in a biological neural cell. (b) Artificial neuronal waveform (spike) emitted by the circuit in panel (a). Taken from Ref. 5.

An even simpler arrangement using a single VO_2 device can be used to create an oscillator circuit. The principle of this is demonstrated in Fig. I.3.3. Take a single VO_2 device, connect it in series with a suitably large resistor, and apply a constant voltage $V_{\text{drive}} = V_0$ (Fig. I.3.3a). According to Fig. I.3.3b, we can plot the illustrative $I(V_{\text{bias}})$ curve of the VO_2 device (red-blue curve) and the $I = (V_0 - V_{\text{bias}})/R_S$ load line (gray dashed line). The latter specifies how much current flows through the circuit at a given V_{bias} . We can see that the load line intersects the memristor's $I(V_{\text{bias}})$ curve in unstable states, i.e., before reaching the intersection of the load line with the HRS's $I(V)$ curve (red circle on the red dotted line), the system switches to LRS at the V_{set} voltage. At this point, the voltage division with R_S causes V_{bias} to decrease, but before the other intersection with the load line would be established in the LRS (blue circle), the system switches back to HRS at V_{reset} . As a result, the memristor will switch back and forth between the HRS and LRS in response to a constant V_0 drive voltage. If we connect a capacitor C in parallel with the memristor (Fig. I.3.3c), we can set the time-scale of the periodic switching between the two states. The such obtained oscillating voltage and current waveform is demonstrated in Fig. I.3.3d. Such oscillator circuit will be established and studied during the second part of the measurement session.

As an interesting application, VO_2 oscillating circuits can be assembled to so-called oscillating neural networks (ONNs), which can solve computational problems through their physical operation, i.e. through the synchronization of the phases. Fig. I.3.4 illustrates this scheme through a recent publication of the researchers of IBM Zürich [6]. The computational problem is map coloring: the countries of South America should be colored such

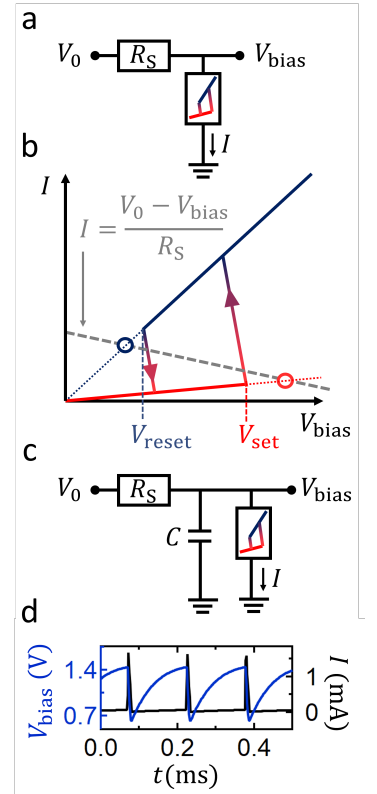


Figure I.3.3. Concept of VO_2 oscillator circuits. (a) A constant V_0 voltage is applied on the VO_2 device and the resistor in series. (b) At sufficiently large R_S the load line does not cross the $I(V)$ curve, i.e. instead of a stable state a repetitive back and forth switching, i.e. an oscillating behavior is established. (c) The characteristic time of the oscillation is adjustable by a parallel capacitor C . (d) Typical voltage (blue) and current (black) signal of a VO_2 oscillator circuit.

that the minimal number of colors are used, but neighbor countries have different colors (Fig. I.3.4a). The map can be represented by a graph whose nodes represent countries, while the connections represent the neighboring positions of countries (Fig. I.3.4b). This graph can be implemented by an oscillating neural network, where the nodes are VO_2 oscillator circuits, and the connections are established by capacitive coupling between the individual oscillator circuits (Fig. I.3.4c). In this network of oscillating circuits the phases of the individual oscillators synchronize such that the phase difference is maximized between neighbor (coupled) nodes. Finally, the phases stabilize around four distinct values: oscillators 1, 2, 3, 6 have fundamentally different phases, while oscillators 4 and 5 have similar phase as oscillator 2 (Fig. I.3.4d). This figure tells us the solution of the map coloring problem, which is illustrated by purple, blue, pink and green colors in Figs. I.3.4a,b.

The above example demonstrates an elegant approach in which physics itself provides the solution through the synchronization of coupled oscillators. It is evident that the exponentially growing computational demands of our time cannot be managed by von Neumann-type computers alone; conventional computing must be complemented with highly energy-efficient application-specific hardware and accelerators. These may operate even at the edge of a network, directly at the origin of data, enabling low-cost, ultra-low-power data analysis. Given the remarkable progress in this field, we can expect that new forms of neuromorphic hardware will soon become part of our everyday devices. As oscillator-based neural networks may emerge as key players in this technological evolution, we should not forget that John von Neumann himself made fundamental contributions to the idea of oscillator networks, which could potentially overcome the limitations of von Neumann architectures [7, 8].

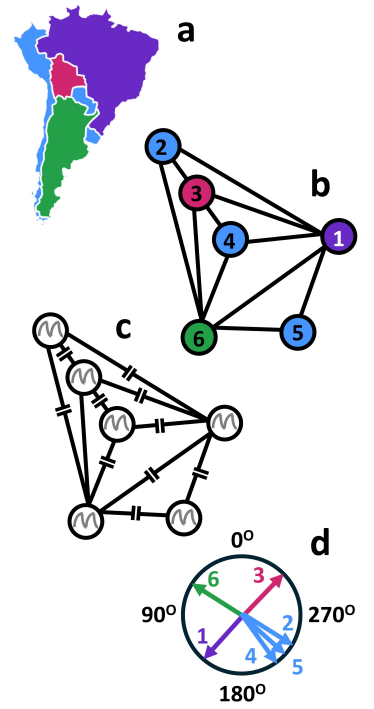


Figure I.3.4. Solving graph coloring problem with the synchronization of a coupled oscillator network. (a) Illustration of the problem: the countries of South Africa should be colored such that the minimal number of colors are used, but neighbor countries have different colors. (b) Graph representation of the problem: counties are represented by nodes, neighbor countries are connected by edges. (c) Mapping of the problem to a network of oscillators. (d) Oscillators synchronize to four different phases corresponding to the solution. Taken from Ref. 6.

Measurement tools

II

List of experimental tools.

- Measurement controller computer with necessary packages installed.
- NI MAX and EasyWaveX softwares installed.
- MyDAQ data acquisition card with BNC connector extension.
- Siglent SDP1168X programmable DC power supply with 230 V supply cable.
- Two USB cables for measurement control.
- Sort BNC cable with T connector.
- Custom box with BNC connectors (see Fig. II.5.1).
- Resistors to be used for the $I(V)$ and $R(T)$ measurements ($330\ \Omega$ and $1\ \text{k}\Omega$).
- 2 SMA female- BNC male adapter.
- 2 high frequency-compatible semirigid cable with SMA connectors (male-male).
- Two magnetic-base micromanipulators with needle probes (GND lines connected with a wire).
- Sample holder with banana connectors (red and black cables for the Peltier temperature control, blue for PT-1000 temperature measurement, see Fig. II.7.1).
- Microscope with installed LED light and external power supply.
- OWON XDS3102A digital storage oscilloscope.
- Siglent SDG 1062X function/arbitrary waveform generator.
- $50\ \Omega$ BNC shunt.
- $1 \times 0.5\ \text{m}$ and $1 \times 1\ \text{m}$ long BNC cables for reference and trigger connections
- $2 \times 0.5\ \text{m}$ BNC cable for connecting the "Box with BNC connectors" into the system.
- Set of capacitors ($100\ \text{pF}$ to $1\ \mu\text{F}$) and resistors ($4.7\ \text{k}\Omega$, $10\ \text{k}\Omega$ and $22\ \text{k}\Omega$) for oscillator measurements.
- Pendrive for data transfer.

1 Measurement setup

In the following, you are going to measure characteristic traits of VO₂ nanogap memristors, produced on chips for this laboratory course. There are 5×5 individual devices located with $500 \mu\text{m}$ spacing on each chip, every memristor having a pair of electrodes with $150 \mu\text{m} \times 150 \mu\text{m}$ surface area (see sample layout later in Fig. II.6.1b-c). To perform AC, DC and temperature-dependent electrical measurements on these samples, a custom-built measurement setup is utilized, shown in the photo of Fig. II.1.1. The following guidelines outline some key considerations for the design of the measurement setup:

- The setup is capable of contacting on-chip VO₂ memristor samples using a custom-built probe station.
- The probe station is designed for the transmission of high frequency signals.
- The sample holder is capable of heating chips above the phase transition temperature.
- Computer controlled measurements can be performed.

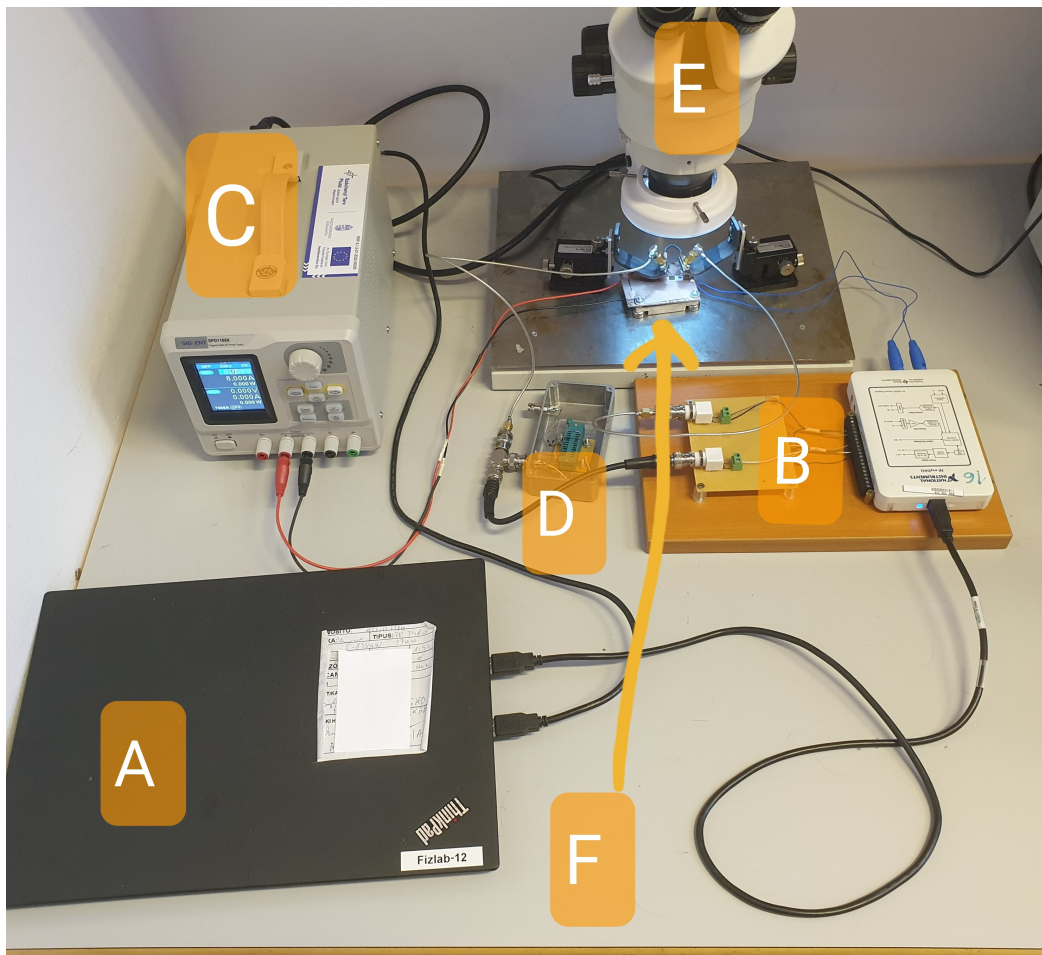


Figure II.1.1. Overview of the tools used for measurements. (A) Computer for measurement control. (B) MyDAQ data acquisition card with holder plate, BNC connectors and USB cable. (C) Siglent SPD1168X programmable DC power supply. (D) Box with BNC connectors for series resistor connection. (E) Microscope with micromanipulators and probes (F) Sample holder with PT-1000 temperature sensor and Peltier temperature control. **Do not touch the sample holder when hot! Be careful with probe needles, as their tips can cause puncture wounds!**

2 Measurement controller computer

Code. // To perform the DC measurements, it is essential to have the following references accessible in C#:

```
Ivi.Visa;
NationalInstruments.DAQmx;
NationalInstruments.Visa;
System;
System.ComponentModel;
System.Globalization;
System.IO;
System.Windows.Forms;
System.Windows.Forms.DataVisualization.Charting;
```

3 MyDAQ data aquisition card

For the DC measurements, a National Instruments myDAQ data acquisition card will be used (depicted with ports and main parameters in Fig. II.3.2). The myDAQ has 16-bit resolution analog differential voltage channels, which have $\pm 2\text{ V}$ and $\pm 10\text{ V}$ range settings. The card also features a 3.5-digit digital multimeter (DMM port), which will be used to measure the resistance of the PT-1000 sensor. The DMM resistance ranges are $200\ \Omega$, $2\text{ k}\Omega$, $20\text{ k}\Omega$ and so on. Detailed specifications and descriptions are available in the [instrument datasheet](#). For easier operation, the myDAQ card is fixed on a board with BNC connectors (Fig. II.3.1).

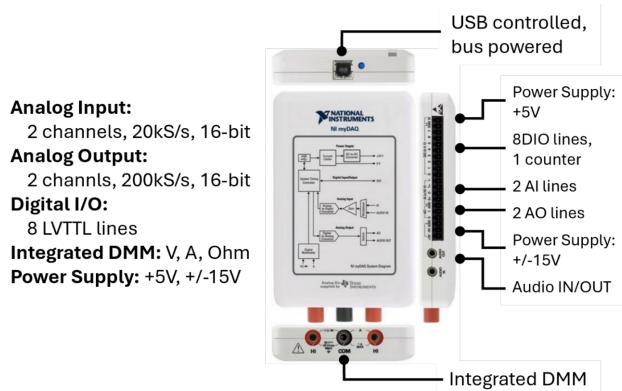


Figure II.3.2. National Instruments myDAQ data acquisition card with its main functionalities and some specifications.

4 Siglent SDP1168X programmable DC power supply

The Siglent SDP1168X programmable DC power supply will be used as a voltage source for the Peltier module. The front panel of the instrument is depicted in Fig. II.4.1. The Peltier module can handle driving voltages up to 12 V and driving currents up to 8 A , but to achieve $T = 68\text{ }^{\circ}\text{C}$ phase change temperature of a VO_2 memristor, typically no more than 8 V driving voltage is needed.

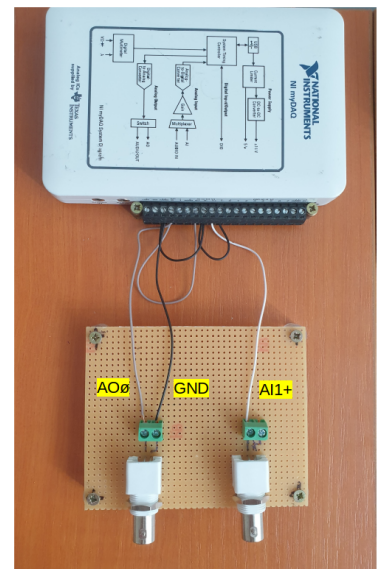


Figure II.3.1. National Instruments myDAQ data acquisition card fixed on a board with BNC connectors for easier operation. myDAQ channels are connected to BNC ports according to the figure.

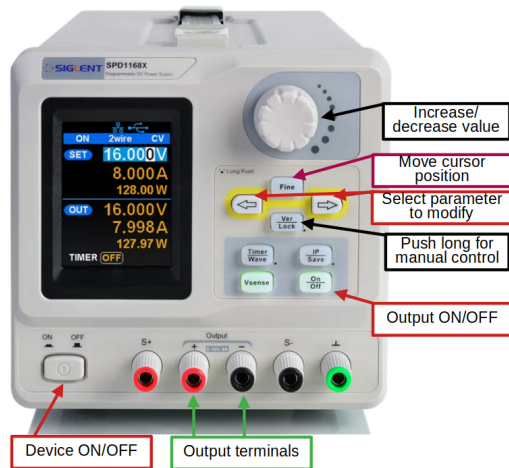


Figure II.4.1. Siglent SDP1168X programmable DC power supply with button descriptions.

Warning. Do not heat up the Peltier module to unnecessarily high temperatures!

The power supply supports remote control via USB or LAN and can be controlled via the Easypower software, LabView or with SCPI commands. To integrate a control for the power supply in your program, use of SCPI commands is recommended via NI VISA protocol. The detailed description of the instrument is available [here](#), while the necessary SCPI commands can be found [here](#) in Chapter "3.4 Command description". To perform $R(T)$ measurements, the commands of output ON/OFF and setting the current and voltage limits should be enough.

Code. // Snippet for power supply identification using NI VISA

```
using Ivi.Visa;
using NationalInstruments.Visa;
//DO NOT FORGET TO ADD THE REFERENCES

Private void button\QueryPowSuplName_Click(object sender, EventArgs e)
{
    string resourceName = SomeDeviceName;
    // Replace with your instrument's resource name
    string command = "*IDN?";
    // SCPI command to query the instrument's identity
    try{
        // Open a session to the instrument:
        NationalInstruments.Visa.ResourceManager().Open(resourceName);
        using (MessageBasedSession session = new NationalInstruments.
            Visa.ResourceManager().Open(resourceName) as MessageBasedSession)
        {
            if (session == null){
                label_QueryPowSuplName.Text = "Failed to open session.";
                return;
            }
        }
    }
```

```

        // Write a command to the instrument:
        session.RawIO.Write(command);
        // Read the response from the instrument:
        string response = session.RawIO.ReadString();
        label_QueryPowSuplName.Text = response;
        session.RawIO.Write("OUTPut CH1,OFF");
    }
}
catch (VisaException ex){
    label_QueryPowSuplName.Text = ex.Message;
}
catch (Exception ex){
    label_QueryPowSuplName.Text = ex.Message;
}
}

```

5 Box with BNC connectors

The R and C elements used for the circuits can be assembled in a sealable metallic box using a zero insertion force socket, in which the pins can be closed by folding down a lever arm. See Fig. II.5.1 for details.

6 Micromanipulators with needle probes

To connect the ≈ 30 nm active region memristors (layout shown in Fig. II.6.1c) into the circuit, each electrodes have $150\text{ }\mu\text{m} \times 150\text{ }\mu\text{m}$ -sized contact pads according to the design shown in Fig. II.6.1b. These contact pads can be connected into the electrical circuit using the needle probes depicted in Fig. II.6.1a. The $50\text{ }\Omega$ impedance matched SMA connector cables are connected to the probes' SMA connectors where the signal line (S) continues

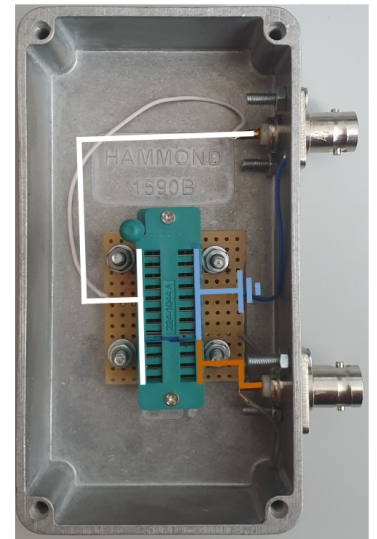


Figure II.5.1. Box with 2 BNC connectors and a 2×12 zero insertion force (ZIF) socket for assembling circuits. Pins marked with white in the image are connected on one side of the ZIF socket, wired to the BNC connector on the left. Pins marked with blue are connected to the ground. Pins marked with brown are connected to the other BNC connector, to the right.

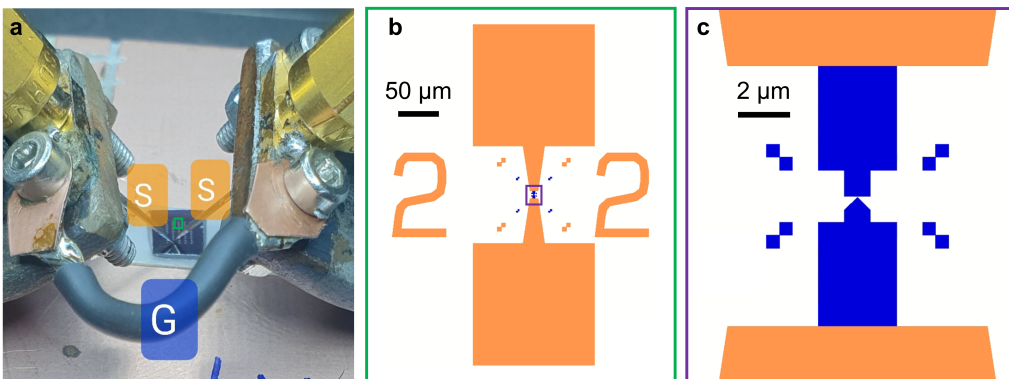


Figure II.6.1. Contacting VO_2 samples with needle probes. (a) Close-up image of a memristor chip mounted on the sample holder. The yellow, hex nut-like metals are the outer parts of the male SMA connectors. Needles (touched to the surface of a memristor's electrode pair) create a secure electrical connection. Signal (S) and ground (G) lines are labeled accordingly. Region highlighted with a green rectangle marks a single memristor device on the chip. (b-c) Layout of a single memristor sample in different magnifications. Panel (b) shows the $150\text{ }\mu\text{m} \times 150\text{ }\mu\text{m}$ electrode pair used for contacting the devices, also showing numbers used for device identification. Region highlighted with a purple rectangle marks the memristor, magnified in panel (c).

through the needles, while the shielding/ground (G) of the coaxial cables are connected via a piece of copper wire. Needles can be moved by turning the screws on the micromanipulators (see Fig. II.1.1), achieving a fine positioning of the needles. If the sample is not in range, coarse positioning can be achieved by gently pushing/sliding the micromanipulators themselves (magnetic force keeps the probes on the stage, but allows in-plane movement). Be careful, do not scratch the surface of the sample, pull up the needles before adjusting the position of the micromanipulators. Also, do not knock the probes over. If you are unsure about moving the probes, ask the lab responsible.

In DC measurements the continuity of the signal and ground lines are enough for a successful measurement (while shielding is advantageous to increase signal-to-noise ratio). In AC measurements, as far as the wavelength of the signals (in our case >3 m) is much longer than the impedance mismatched needle probes, this can be handled as a lumped (not distributed) impedance mismatch. At higher frequencies (>100 MHz) the wave nature of the electric signals has to be considered and special waveguide design probes and samples have to be used. For our measurement frequencies ($\lesssim 30$ MHz), the approximately same signal and ground wire length in the probe region is enough to get minimal signal distortion. In the beginning of the high frequency measurements you will test the distortion and loss of the probes.

Warning. Tip of the needle probes can cause serious injuries. Handle the probes with extra care, do not remove the micromanipulators from their steel base.

Warning. Needles are sensitive to high forces, so do not move the sample holder or the micromanipulators on the steel base when the probes are in low position or even touching the sample.

To make contact, set the microscope magnification to minimum. Then move the probes on the metal plate over the surfaces to be contacted, while the probes are in the upper position. Check that the contact point is in the center of the microscope's field of view and that the tips of the needle probes are also in the center of the field of view. If not, move the micromanipulators on the metal plate and check that their x - y position is in the middle. Increase the magnification to approximately $3\times$, then focus on the point of the surface to be contacted. Use the z position adjustment screw to move the needle until its tip is in focus. Then move the needle closer to the surface to be contacted until the tip moves minimally in the x - y direction. If the needle is not in focus, check with bare eye that there are no mechanical obstacles in the path of the probe.

7 Sample holder with Banana connectors

In Fig. II.7.1 the sample holder is depicted with the Peltier module connector cables, the VO₂ chip, a thin film technology SMD reference resistor and a PT-1000 temperature sensor. While the Peltier module is responsible to change the temperature of the VO₂ chip, the achieved temperature can be recorded using the PT-1000 thermometer, whose resistance dependence is as follows:

$$R[\Omega] = 1000 \, \Omega + 3.851 \, \frac{\Omega}{^{\circ}\text{C}} \cdot T[^{\circ}\text{C}].$$

The connectors of the PT-1000 thermometer can be connected to the ports of the myDAQ's DMM using the blue banana cables, while the red and black banana cables are connected

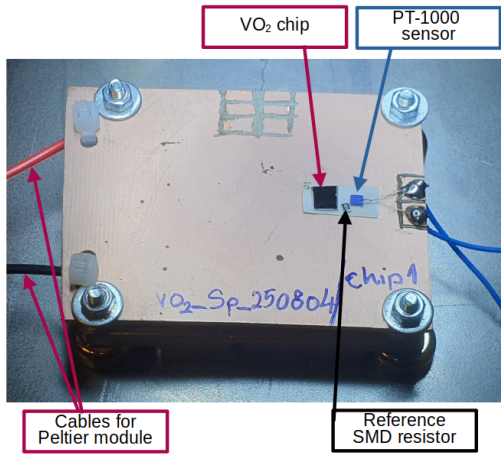


Figure II.7.1. Sample holder with the Peltier module connector cables, chip with VO₂ memristors, a reference resistor and a PT-1000 temperature sensor.

to the Siglent power supply to drive the Peltier module. In the latter case, selecting the correct polarity is crucial; make sure to connect the red and blue cables to the positive and negative outputs, respectively.

8 Siglent SDG 1062X function/arbitrary waveform generator

The Siglent SDG 1062X arbitrary waveform generator (AWG) depicted in Fig. II.8.1 will be used to generate arbitrary waveforms for fast switching measurements. It features a maximum bandwidth of 60 MHz and a peak-to-peak output amplitude of up to 20 V while operating at 150 MSa/s sampling rate. We aim to get the shortest FWHM pulses (preferably < 30 ns) for fast switching, followed by a longer (200 ns) readout offset level. It would be common sense to operate the AWG at the highest sampling rate and have only 1 data point defined as the set pulse. Sadly, the DDS playback mode (featuring the shortest FWHM) occasionally misses one data point at 150 MSa/s, therefore we will operate the AWG at 30 MSa/s instead. Since we define only 1 datapoint for the switching pulse, 25 ns FWHM pulses are still achievable.

The device provides standard connectivity options, including USB Host, USB Device (USBTMC), and LAN (VXI-11). Instead of controlling this device with SCPI commands,



Figure II.8.1. Siglent SDG 1062X function/arbitrary waveform generator.

	A	B
1	data length	64
2	frequency	3000000
3	amp	10
4	offset	5
5	phase	0
6		
7		
8		
9		
10		
11		
12		
13	xpos	value
14	1	0
15	2	0.25
16	3	0.25
17	4	0.25
18	5	0.25
19	6	0.25
20	7	5
21	8	0.25

Figure II.8.2. A snippet of the waveform file used for the relaxation time measurements. "Data length" stands for the number of "xpos" values, always use 64 data points for the best performance from the AWG. "Frequency" defines the length of the whole waveform. Set it to 30 MHz. "Amp" and "offset" values should be set according to the highest voltages you want on the output. These values only define the range. The "xpos" and "value" pairs define the waveform shape. First and last points should be zero. For the smallest FWHM pulses, define only one point as the programming voltage.

the EasyWaveX software will be used to upload arbitrary waveforms to the function generator (screenshot with main functions is shown in Fig. II.8.3). After downloading EasyWaveX to a measurement controller computer, explore the example waveform files provided with the program. An example waveform file can be found [here](#), while a snippet with the description of how the waveform file is structured can be seen in Fig. II.8.2. Instead of continuously repeating the waveforms during the measurements, a manually triggered bursting mode will be used, to protect the samples from degradation. The trigger output can be found on the back panel.

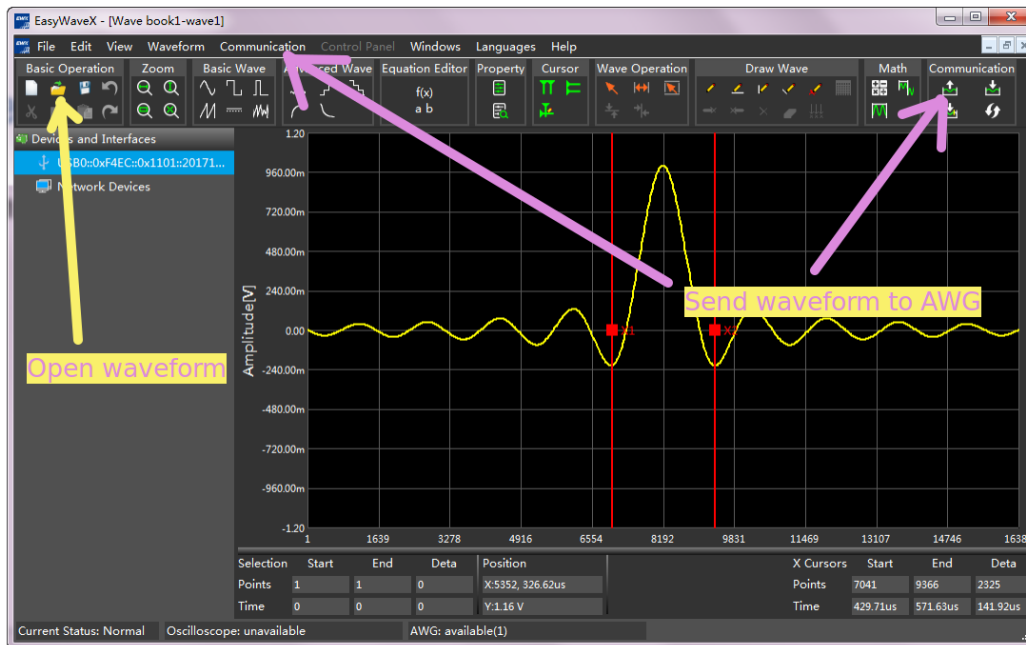


Figure II.8.3. EasyWaveX program for 1062X function/arbitrary waveform generator.

9 OWON XDS3102A digital storage oscilloscope

An OWON XDS3102A digital storage oscilloscope is used to measure the transmitted and reference signal in the high frequency measurements. The major functions of this oscilloscope are summarized in Fig. II.9.1. For detailed specifications, further information and illustrative videos, visit [OWON's web site](#).

To record the reference waveform used for switching the VO₂ memristor, together with the transmitted signal, both channels of the oscilloscope (OSC) are used, while the external trigger input ensures synchronization to the AWG. To avoid unwanted reflections, 50 Ω impedance matched OSC input channels are desired. This can be achieved by connecting the input signals to the HighZ OSC inputs with T-connectors and placing BNC 50 Ω resistors on the other sides of the T-connectors.

Due to a bug in the oscilloscope's software, after saving a waveform to a USB stick (External storage) you have to move the TIMEBASE knob back-and-forth to revive the software of the oscilloscope.

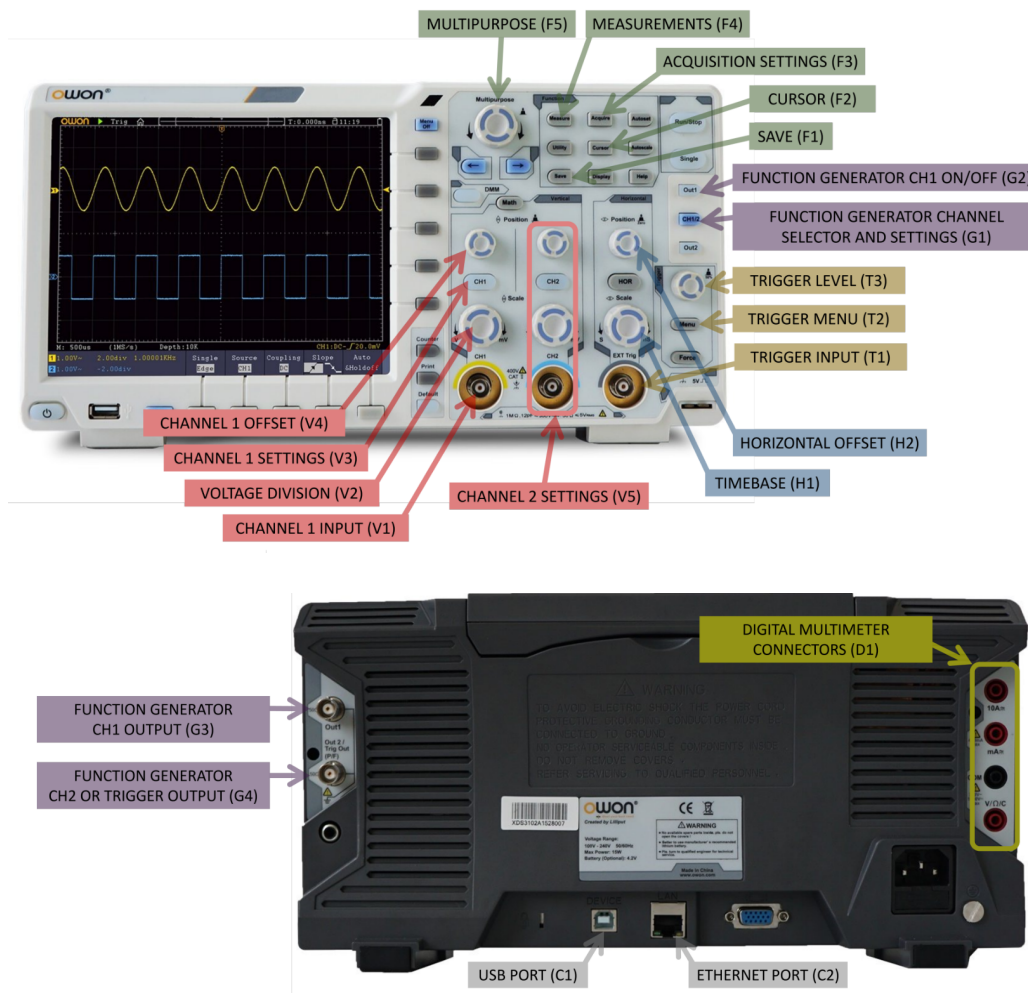


Figure II.9.1. Top: Front cover photo of the OWON XDS3102A digital oscilloscope, showing the main connectors and control buttons. The vertical menu (V1-V5) is used for setting up the voltage inputs. The V2 button determines the voltage range for the first input channel (V1), defining the voltage corresponding to one vertical division on the screen. V3 adjusts the settings of channel 1 input, and V4 allows vertical offsetting of the measured waveform. Similar functionalities are available for the second voltage input (channel 2, V5). The horizontal menu (H1-H2) manages temporal settings, with H1 defining the time duration of one horizontal division and H2 shifting the measured waveform horizontally. The trigger system (T1-T3) ensures signal synchronization. While the T1 trigger input channel can sync waveforms to an external trigger source, synchronization to CH1 or CH2 channels is also possible. T3 sets the trigger level, with more delicate settings available in the trigger menu (T2). Further functions (F1-F5) include saving (F1), adjustable cursors for manual measurements (F2), acquisition settings (F3), built-in frequency and amplitude measurements (F4) and a multipurpose knob (F5). The settings of the built-in function generator (G1-G4) can be accessed with the "CH1/CH2" button (G1). **Bottom: Back cover of the OWON XDS3102A digital oscilloscope,** showing the function generator outputs. (G3-G4) can be switched on/off with the "Out1/Out2" button. The back of the oscilloscope also includes USB and Ethernet connectors for communication (C1-C2) and a built-in digital multimeter (D1).

Measurement Tasks

Task 1. Build the measurement circuit and the control program for the $I(V)$ measurement, and test it on a reference SMD resistor.

Measurement Methods. Memristors are usually characterized by driving them with a voltage signal while measuring the current. A typical waveform used for characterization is depicted in Fig. III.1.2. To measure the $I(V)$ curve of the volatile VO₂ memristor, the circuit shown in Fig. III.1.3 is used. With the myDAQ data acquisition card, the circuit can be driven with a variable voltage between -10 V and 10 V, which is sufficient to reach the switching threshold value V_{set} of the nanogap VO₂ memristor by selecting the appropriate series resistance. In the measurement circuit (see Fig. III.1.3), the voltage driving is accomplished by using the myDAQ AO0 channel. The card has a maximum current output of 2 mA, which is quite low and might be reached during measurement. This can occur when the memristor switches to a LRS with especially low resistance and the load on the entire system is reduced to practically $\approx R_S$. If the 2 mA current limit is violated, the programmed control voltage cannot be achieved. To recognize this, it is essential to monitor the actual voltage output using the myDAQ AI0 channel. The current flowing through the system is calculated from the voltage drop across the series resistor, measured on channel AI1, and the known series resistance R_S . The bias voltage V_{bias} across the memristor is given by the difference between the voltages measured on channels AI0 and AI1. The memristor samples are contacted and connected into the circuit using needle probes mounted on micromanipulators (see Figure II.6.1). The ground line of the probes are connected via a short wire while the signal line is connected through the probe needles touched to the surface of the sample's electrodes.

1. Discuss the operation of the probe station with the measurement supervisor and practice using it by contacting the two ends of the 4.7 k Ω SMD reference resistor on the sample holder plate with the probe station needles. Check the resistance with a handheld multimeter. Keep the needles on the SMD resistor until the end of this task. See Section II.6 for the operation of the probe station.

Warning. Handle the probe needles with extra care, probe tips can cause serious injuries! Also, use the micromanipulators carefully, since too much force applied on the needles can break them.

2. Assemble the measurement circuit schematically depicted in Fig. III.1.3 and also shown in Fig. II.1.1, but instead of the sample VO₂, use the reference SMD resistor described above. Use a series resistance of 2 k Ω , which can be installed to the box with the BNC connectors (Fig. II.5.1).
3. Connect the NI myDAQ via USB to the measurement controller computer. An example code in C# is provided [here](#). You have to extend the measurement part and write a few functions for the code. Based on the following tasks, think it over what

III

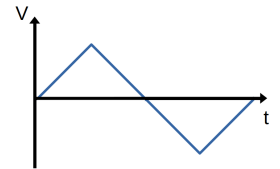


Figure III.1.2. A triangle wave provides a continuous rate of change in the driving voltage, which is ideal for recording $I(V)$ traces like those depicted in Fig. I.3.1.

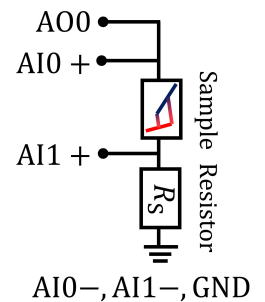


Figure III.1.3. Circuit schematics for the $I(V)$ characterization of memristor samples highlighting the input/output channels of the myDAQ. The same circuit is used for the $R(T)$ measurements.

structure is the best for your purposes, **then** start coding. Check in NI MAX what is the myDAQ device's name, this will be used in your C# program.

Code. A step-by-step guide on how to build up your code:

- (a) Sweep the voltage on the AO0 channel of the myDAQ back-and-forth using a timer event.
- (b) Measure the real output voltage and the voltage drop on the series resistor using myDAQ AI0 and AI1 channels.
- (c) Measured points should be plotted and written into a file at each clock tick. Required data to save: time, drive voltage and voltage drop on the series resistor (raw data); bias voltage and current of the memristor (evaluated data). Required plot: $I(V_{\text{bias}})$ curve of the device under study.
- (d) Make the sweep configuration parameters accessible externally, from the user interface.

4. Measure the $I(V_{\text{bias}})$ curve of the SMD resistor, evaluate the resistance from the slope. Compare it to the nominal resistance value.

Task 2. Measure the $I(V)$ characteristics of a VO_2 memristor.

Warning. Do not contact the VO_2 memristor into the circuit until you tested every functionality of your program and got an approval from the lab supervisor.

1. Connect a VO_2 sample into the circuit using the probe station. [Check the online tables for functional devices on your chip. If one device is not functional anymore on the chip, make a note in the table.](#) See Sec. II.6 for the details of the probe station.

Warning. Ask the lab instructor to supervise this activity continuously. Improper use of the probe station can completely destroy the probe station needles and the chip containing the samples.

2. Perform an $I(V)$ measurement with a properly chosen series resistor. Note, that at too low series resistance the 2 mA current limit might be violated, while a too high series resistance would yield an oscillating behavior instead of a conventional $I(V)$ curve with hysteretic switching.

For future reference, read the set and reset voltages from the measured $I(V)$ curve.

Warning. To ensure the safety of the sample, do not increase the drive voltage above 4 V. If no switching is detected up to this signal level, ask the lab instructor for help.

Task 3. Measure the $R(T)$ characteristics of a VO_2 memristor.

Measurement Methods. To measure the $R(T)$ dependence of the VO_2 memristor, the same electrical circuit is used as for the $I(V)$ measurement, but now the driving voltage is kept constant. The temperature is controlled by a Peltier module. The latter is built inside the sample holder, which carries the memristor chip on its top surface (see tool F in Fig. II.1.1 and also Fig. II.7.1 for a close-up image of the sample

holder). The Peltier module creates temperature difference between the two sides of the cell when voltage is applied. Here, the side facing the sample will be heated when positive voltage is applied to the red cable of the Peltier module. An SDP1168X programmable DC power supply (see tool C in Fig. II.1.1) serves as a voltage source for the Peltier module. To measure the temperature of the sample holder, a PT-1000 sensor is used, which changes its resistance according to its temperature variation. Since neither side of the Peltier module is connected to a fixed temperature surface, it is not recommended to write a PID control for temperature sweep, instead, just sweep the supply voltage of the Peltier module slowly until the resistance of the memristor drops. The voltage sweep of the power supply should be slower than 0.1 V/15 sec in order to have approximately the same temperature on the PT-1000 sensor and the sample, and also to have enough data points at each temperature.

1. Connect the Siglent SDP1168X power supply to the computer, and extend your program code with the $R(T)$ measurement functionality according to the items listed below. Check in NI MAX the USB address of the Siglent power supply, this will be used in your C# program.

Warning. The sample, Peltier element and thermometer wires should be disconnected during programming. Only after ensuring that the program is functioning properly and with the approval of the lab leader should you connect these devices.

Code. A step-by-step guide on how to build up your code:

- (a) Create an input box for the applied constant drive voltage on the user interface.
- (b) Control the voltage supply using VISA and SCPI commands, sweep the voltage back-and-forth (check out Sec. II.4).
- (c) Measure the resistance of the PT-1000 sensor using myDAQ DMM channel.
- (d) Determine the sample resistance from the measured AI0 and AI1 voltage values.
- (e) Measured points should be plotted and written into a file at each clock tick. (Required data to save: time, voltage on the Peltier element, PT-1000 resistance, drive voltage on the sample, voltage drop on the series resistor, temperature, sample resistance.)
- (f) Make the sweep configuration parameters accessible externally, from the user interface.
- (g) Make it possible to switch the power supply output ON and OFF from the user interface.

2. Record the $R(T)$ curve of the VO₂ device during heating and cooling. Read the phase transition temperature and compare it with the value given in the literature.

Task 4. Measure the relaxation time (reset time) of the VO₂ memristor after a programming (set) pulse.

Measurement Methods. The voltage variation of a capacitance discharged through a resistor has a well-defined time constant, regardless of the initial charge placed on the capacitance. This is called exponential relaxation. However, the relaxation of VO₂ memristors from a LRS to a HRS has complex, non-exponential dynamics. Under optimized conditions, this reset transition is shorter than 1 ns [9], but after a high-amplitude programming pulse or at a relatively high readout voltage, the relaxation into the HRS can last even three orders of magnitude longer, on the order of microseconds. This task provides insight into this striking non-exponential relaxation dynamics. Note, that sufficiently rapid relaxation is essential for the fast and energy-efficient operation of oscillatory neural networks.

The *Wave phenomena in transmission lines* measurement taken during the *Measurement Design and Laboratory Exercises* course showed that high-speed measurements require a different approach than conventional DC measurements. In the former case, the wave propagation in the transmission lines is crucial, and a resistor in a transmission line behaves similarly to a semi-transparent optical mirror: it reflects a part of the incoming signal wave while allowing the other part to pass through. This situation is illustrated in Fig. III.4.1a. An arbitrary waveform generator (AWG) outputs a driving pulse sequence $V_{IN}(t)$, which propagates towards the resistor R as an incoming wave. A part of this wave is transmitted through the resistor, and propagates to an oscilloscope (OSC), where $V_{TRANS}(t)$ is measured. It was shown in the *Wave phenomena in transmission lines* measurement, that the ratio of the transmitted and incoming wave is

$$\frac{V_{TRANS}(t)}{V_{IN}(t)} = \frac{2Z_0}{2Z_0 + R}, \quad (\text{III.4.1})$$

where $Z_0 = 50 \Omega$ is the wave impedance of the transmission lines, R is the resistance of the in-line resistor, and the incoming and transmitted signals are aligned in time. It is noted, that the AWG and the OSC should have 50Ω output and input resistances to avoid reflections. At the OWON OSC the 50Ω termination is not a built-in option, so it is realized by placing a 50Ω resistor to the input with a T-piece. It is noted, that for a purely resistive in-line circuits element Eq. III.4.1 is not only valid for harmonic waves, but for an arbitrary waveform, i.e. the transmission through the resistor preserves the waveform, just modifies the amplitude.

This scheme can be also utilized to follow the resistance variation of a memristive element, $R_M(t)$ in real time. An incoming pulse is sent to the memristor, the transmitted pulse is measured, and the resistance variation is extracted from the equation

$$R_M(t) = 2Z_0 \cdot \left(\frac{V_{IN}(t)}{V_{TRANS}(t)} - 1 \right), \quad (\text{III.4.2})$$

where $R_M(t)$ is expressed by the reorganization of Eq. III.4.1. Utilizing this scheme the relaxation dynamics of the VO₂ device can be traced as follows. The device is driven by a pulse sequence, which includes a programming pulse and readout voltage levels before and after it (Fig. III.4.1b). The first readout level probes the resistance of the initial HRS, the programming pulse switches the memristor to the LRS, and the second readout level traces its relaxation towards the HRS. At the beginning of the second readout level the memristor is in its LRS, so Eq. III.4.2 yields $R_M = R_{LRS}$. Af-

ter a sufficient amount of time, the memristor returns to its HRS, so Eq. III.4.2 yields $R_M = R_{HRS}$. Between the two, based on Eq. III.4.2, we can follow the relaxation of $R_M(t)$ in real time. This principle is summarized in Fig. III.4.1c,d,e.

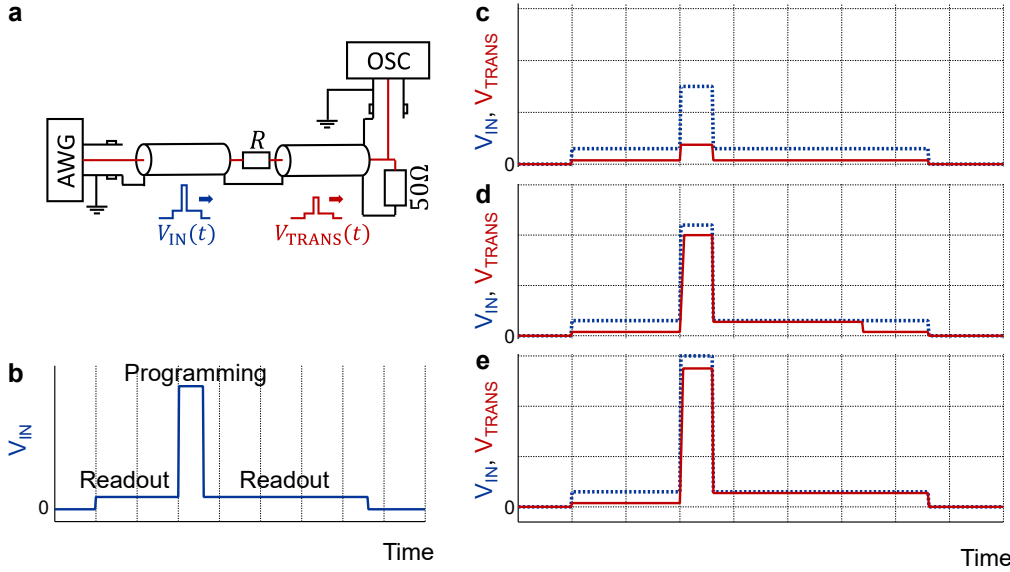


Figure III.4.1. Concept of fast, time-resolved relaxation time measurements. (a) Scheme of the measurement. The memristor sample behaves as a resistor, and is connected to coaxial cables at the two sides. At one side, the sample is driven by a pulsed waveform from an AWG, while on the other side the signal transmitted through the sample is fed to an OSC. To avoid reflections, both the AWG output and the OSC input are $50\,\Omega$ terminated, in the latter case this is achieved by an external $50\,\Omega$ resistor. (b) Scheme of the driving sequence. A programming pulse is initiated and followed by a lower readout voltage level, otherwise the signal is zero. (c) An illustrative example for a situation, where the programming pulse amplitude is not high enough to induce a set transition. In this case, the memristor resistance is instantly $R_M(t) = R_{HRS}$, i.e. the transmitted – incoming signal ratio is $V_{TRANS}/V_{IN} = 100\,\Omega/(100\,\Omega + R_{HRS})$. (d) A higher amplitude programming pulse induces a set transition, and right after the programming pulse $R_M(t) = R_{LRS}$ is satisfied. At the end of the second readout segment $R_M(t) = R_{HRS}$ is satisfied, and the reset transition happens inbetween. The variation of $R_M(t)$ is deducible from Eq. III.4.2. Note, that here a stepwise LRS to HRS transition is illustrated, while in the measurement a gradual one will be observed. (e) An even larger pulse amplitude the reset time elongates such that the relaxation is not observable until the end of the second readout segment.

1. Assemble the measurement setup. CH1 of the AWG is connected to one side of the probes, while the other probe SMA connector is connected to CH1 of the OSC, where a $50\,\Omega$ resistor on a T-piece grants impedance matching. CH2 of the AWG goes directly to CH2 of the OSC with similar $50\,\Omega$ termination. Synchronization is achieved through the TRIG OUT and TRIG IN channels of the AWG and OSC. The output waveform of the AWG is defined and uploaded by the Siglent EasyWaveX software. Make sure to apply the same waveform on the both channels of the AWG, so that CH2 is the same as the driving signal of the sample.
2. Set both channels of the AWG to burst mode, set trigger to manual (both channels). Don't forget to set trigger out to UP on one of the channels. Use "normal" trigger on the oscilloscope. **Test the signal loss through the probe station. For this, connect both needle probes to the same contact pad of the $4.7\,\text{k}\Omega$ SMD resistor using the micromanipulators. Small movement of the SMD resistor upon contacting is normal, but too much movement indicates too much force. Record the reference waveform**

and the transmitted one. Determine the loss through your probes. If it is above 3 dB (below $\approx 70\%$ in voltage ratio), consult the lab responsible.

- Next, connect the needle probes to the **different contact pads** of the $4.7\text{ k}\Omega$ SMD resistor using the micromanipulators, and apply the signal. Check whether the evaluation of the measurements through Eq. III.4.2 provides reasonable value for the resistance (don't forget to compensate with your probe loss).
- Investigate the switching dynamics of the device. First, install a waveform on the AWG that is approved by the lab leader. Note that a programming pulse that is too low is not sufficient to activate the switching, while one that is too high may destroy the sample. The readout level must be selected so that it can be measured accurately without causing a switching.

Contact a VO_2 sample with the micromanipulators. If the sample does not switch to the LRS, increase the set pulse level in 25% steps. Save every waveform you produce. Continue adjusting the driving pulse shape until you reach a situation, where (i) the programming pulse switches the device from the HRS to the LRS, and (ii) the relaxation from the LRS to the HRS is clearly visible along the second readout segment. Evaluate this waveform, specify an upper limit for the characteristic time of the set transition, and determine the characteristic time of relaxation (reset process). If you have enough time, test the dependence of the reset time on the amplitude of the programming pulse and the readout level.

Task 5. Assemble and investigate a VO_2 oscillator circuit.

Measurement Methods. Oscillatory measurements can also be performed using the above AWG and OSC instruments. In higher frequencies, high impedance voltage measurement inputs with long coaxial cables can induce unwanted reflections, therefore instead of measuring V_{bias} as depicted in Fig. I.3.3c, we measure the current flow according to Fig. III.5.1. The $50\ \Omega$ shunt applied on the input of the OSC is perfect for this purpose, since measuring the voltage drop on it, the current can easily be calculated. To avoid unwanted reflections in the circuit, it is essential to use as short cables as possible. To this end, use the shorter black SMA terminated coaxial cables instead of the longer metallic semirigid ones to connect the sample to the BNC box, which includes the series resistor R_S and the parallel capacitor C .

To avoid sample degradation, it is useful to restrict the number of oscillations studied. To achieve this, instead of a continuous finite V_0 drive voltage, the circuit must be driven by a single, manually triggered burst that outputs the V_0 voltage for a shorter period of time and otherwise applies zero voltage.

- Assemble the circuit and set up the burst mode of the AWG. Initially use a parallel capacitance of $C = 1\text{ nF}$. Relying on the set and reset voltage values, find a proper series resistance value, which satisfies the criteria in Fig. I.3.3b.

Warning. Ask the lab instructor to approve your circuit and settings, do not apply the driving signal to the sample prior to that.

- Trigger a burst, and check, whether oscillations are observed on the OSC. If not, modify the amplitude of the burst. Record an oscillating pattern, and determine its frequency by the OSC cursors.

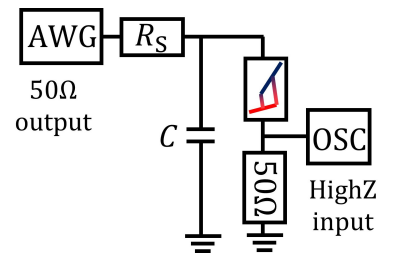


Figure III.5.1. Rearrangement of the circuit shown in Fig. I.3.3c for high frequency oscillatory measurements. C and R_S are installed in the BNC box (see in Fig. II.5.1). Minimize the cable lengths for a successful measurement.

3. Investigate the dependence of the oscillation frequency on the driving voltage amplitude and the parallel capacitance C .

Some examples for the entrance test questions:

1. How do you calculate the resistance of the memristor in the pulsed measurements (formula)?
2. How do you calculate the resistance of the memristor in the DC measurements (formula)?
3. What type of memristors do you know?
4. How do you control the temperature during the measurement?
5. Where can memristors be applied, if we consider energy efficiency?

Contributions

The measuring setup was developed and assembled by Zsigmond Pollner, with the collaboration of András Halbritter (conceptualization and contribution to the laboratory description) and Tímea Török (sample preparation, testing, and contribution to the laboratory description). The VO₂ memristor samples were manufactured in the Nanosensors Laboratory of the Institute of Technical Physics and Materials Science (HUN-REN Centre for Energy Research).

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